Quartic-extension genus-5 cover: toy reproduction and end-to-end measurement

Study `research/humbert_genus5_cover_20261021` · runs 2026-10-21 to 2026-10-26 · PR #2580

Scope. Toy fields only: q prime with $17 \le q \le 8009$, E over $\mathbb{F}_{q^4}$, prime subgroup order $r < 2^{62}$ with cofactor at most 128. All computation is native Rust in `src/` with no dependencies; `summarize.py` and `plots.py` only lay out recorded values. Each statement is labelled **verified** (an exact check that runs on every instance), **measured** (a timed or counted run; records in `results/`) or **derived** (argued in the code comments of `src/cover.rs`, not separately proved here).

Host. 4-vCPU Intel Xeon @ 2.10 GHz shared cloud VM, no CPU isolation, no hardware counters. Index calculus and rho for an instance run back to back in the same process, so each ratio is paired; absolute times carry host noise.

1. Lead 1: reproduction of the family

The descended curve C is built as an explicit model over $\mathbb{F}_q$ (a twisted Humbert curve, `src/cover.rs`). `humbert verify` counts $N_{k(C)}$ for $k = 1..5$ on that model alone, forms the L-polynomial, and compares it with $L_{E(T^4)} \cdot L_{D(T)}$, where $\#E(\mathbb{F}_{q^4})$ comes from baby-step giant-step on E and $\#D(\mathbb{F}_q)$ from the extra factor's own equation $D : y^2 = N(\eta)m_\alpha(x)$.

instances	L-poly match	twisted-E control rejected	$\#\mathrm{Jac}(C)(\mathbb{F}_q) = \#E \cdot \#D$	h_1-linear form = Möbius cubic	q
12	12/12	12/12	12/12	12/12	17, 19, 23; both twists; 2 seeds

Table 1: Verification grid (`results/verify.jsonl`). All ten L-polynomial coefficients agree on every instance: genus 5, defined over $\mathbb{F}_q$, $\mathrm{Jac}(C) \sim \mathrm{Res}\ E \times D$.

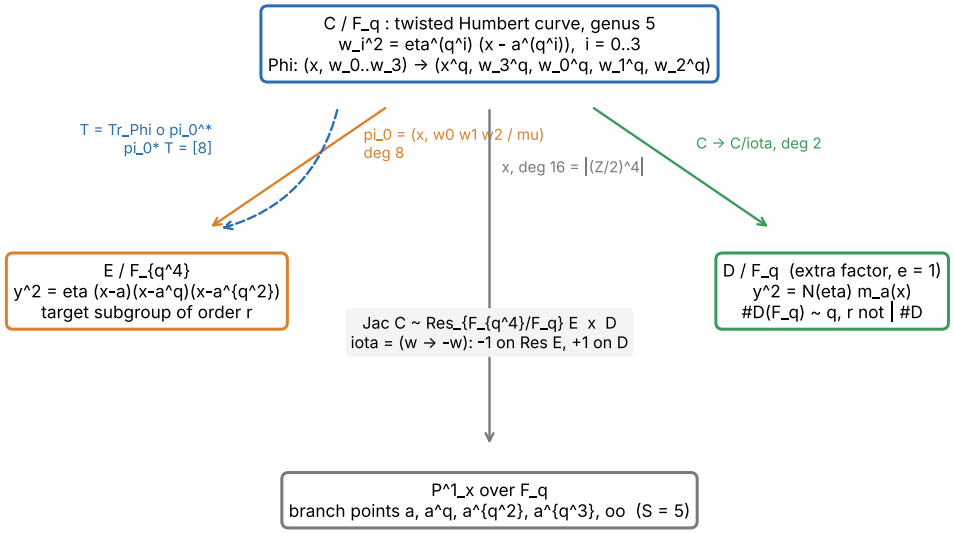


Figure 1: Objects and maps of the construction (`src/cover.rs`); no data.

Transfer (verified on every attack run). The pushforward of the transferred divisor $T(P)$ to E is computed point by point and equals $8P$; hence $T(P) \neq 0$ for the generator of the order-r subgroup. The discrete log is then recovered by index calculus on C through T: **90/90 runs correct.**

2. Lead 2: binary-pattern classifier

`humbert classify` enumerates Frobenius-orbit root patterns for $d \in \{2, 3, 4\}$ and source genus 1..3 (11,200 patterns, `results/classify.jsonl`). On every pattern the Riemann–Hurwitz genus equals the independent Kani–Rosen sum. The quoted family appears in both forms ($\deg h_1 \in \{0, 1\}$) with $m = 4$, $S = 5$, $g = 5$, $e = 1$, both twists allowed, non-hyperelliptic, one extra genus-1 factor over $\mathbb{F}_q$. For $d = 4$ the minimum excess genus is $e = 1$ for every source genus ($g = 5, 9, 13$). For $d = 2$ and $d = 3$ the classifier reproduces the known $e = 0$ covers (genus 2 hyperelliptic; genus 3, partly non-hyperelliptic). **Not done:** the remaining Appendix A rows were not compared, because the paper text was not available in this session.

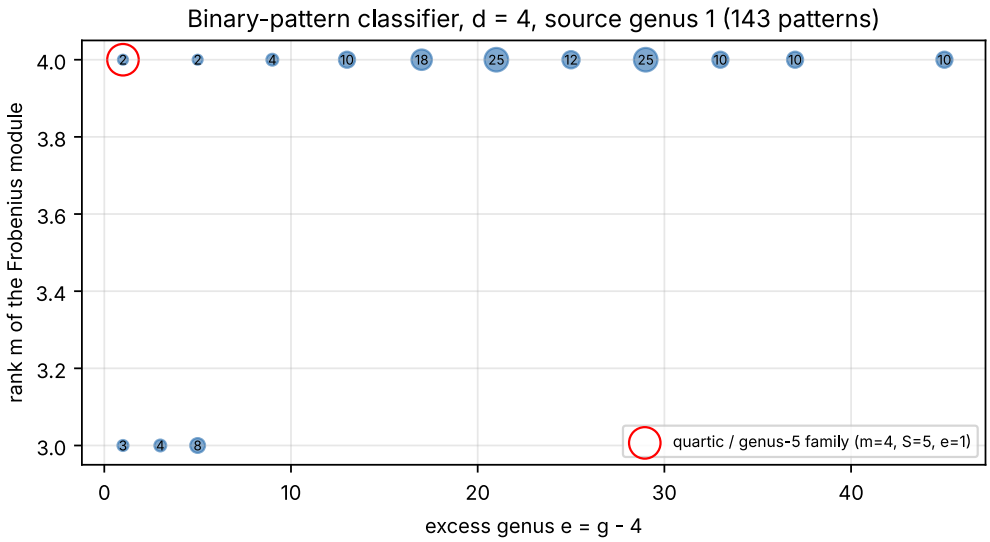


Figure 2: Patterns for $d = 4$, source genus 1, by excess genus and module rank (`results/classify.jsonl`).

3. Lead 3: complete attack against matched Pollard rho

Grid: $q \in \{61, \ldots, 8009\}$ (15 primes), 3 seeds, two arms (full factor base; factor base halved by the involution ι). Phases recorded per run: cover set-up, relation search, linear algebra (dense elimination up to 1200 unknowns, scalar Wiedemann above), individual-log descent for P and Q, peak RSS. Rho is a 32-adding walk with Brent cycle detection on the same subgroup. Two rho references are shown: the measured single run, and the expected-steps time $\sqrt{\pi r/2}$ times that run's measured seconds per step (lower variance).

q	$\log_2 r$	arm	unknowns	relations	rel. s	LA s	descent s	IC s	rho meas. s	rho exp. s	IC / rho exp. (range)
61	21.7	full	64	77	0.096	0.001	1.359	1.46	0.003	0.004	887.57 [381.85, 983.44]
61	21.7	iota	33	42	0.085	0.000	1.348	1.44	0.002	0.003	777.56 [445.04, 1181.98]
101	21.6	full	98	103	0.246	0.001	1.216	1.47	0.004	0.003	649.73 [226.20, 1039.00]
101	21.6	iota	50	55	0.178	0.000	1.356	1.54	0.004	0.003	546.97 [265.11, 1251.71]
151	23.4	full	154	159	0.275	0.006	0.638	0.90	0.007	0.004	203.31 [143.23, 482.38]
151	23.4	iota	78	83	0.162	0.001	0.539	0.71	0.007	0.005	138.93 [74.83, 520.14]
211	27.3	full	204	209	0.390	0.011	0.574	0.99	0.010	0.012	84.44 [79.16, 148.22]
211	27.3	iota	103	108	0.238	0.002	0.492	0.75	0.018	0.012	76.63 [50.06, 112.69]
307	27.6	full	292	297	0.561	0.028	0.492	1.21	0.009	0.025	48.52 [39.49, 92.84]
307	27.6	iota	147	152	0.365	0.005	0.497	0.88	0.035	0.012	66.36 [34.05, 99.27]
401	30.3	full	426	431	0.588	0.091	0.421	1.11	0.046	0.035	37.11 [13.16, 60.58]
401	30.3	iota	214	219	0.316	0.009	0.322	0.66	0.017	0.030	21.52 [10.70, 34.94]
503	29.3	full	498	503	0.924	0.149	0.481	1.78	0.043	0.024	69.01 [56.53, 73.72]
503	29.3	iota	250	255	0.407	0.019	0.438	0.89	0.022	0.023	40.33 [23.61, 46.68]
701	34.2	full	680	685	1.301	0.366	0.524	2.17	0.171	0.110	19.71 [10.70, 28.64]
701	34.2	iota	341	346	0.692	0.037	0.541	1.25	0.128	0.113	13.37 [5.66, 16.06]
1009	34.0	full	1048	1053	1.656	1.120	0.219	3.10	0.107	0.122	27.46 [6.54, 36.46]
1009	34.0	iota	525	530	0.851	0.169	0.360	1.44	0.303	0.111	12.99 [3.55, 15.26]
1511	38.2	full	1488	1495	2.600	1.806	0.509	4.91	1.636	0.536	9.62 [8.89, 17.60]
1511	38.2	iota	745	750	1.353	0.375	0.403	2.15	1.252	0.548	3.92 [2.70, 7.19]
2003	39.9	full	2020	2059	3.785	3.366	0.391	7.57	0.759	0.855	8.85 [5.95, 23.92]
2003	39.9	iota	1011	1033	1.737	1.261	0.540	3.62	1.758	0.862	4.00 [3.32, 10.10]
3001	40.3	full	2924	3221	6.601	21.854	0.279	28.97	2.358	0.892	16.10 [7.19, 59.59]
3001	40.3	iota	1463	1468	3.080	1.969	0.256	5.77	1.693	1.028	6.10 [1.32, 15.16]
4001	45.9	full	3964	4167	7.561	25.664	0.452	33.70	4.358	6.857	15.29 [4.86, 16.44]
4001	45.9	iota	1983	1988	3.662	3.471	0.309	7.91	5.698	6.760	1.14 [1.06, 6.60]
6007	45.2	full	6046	6353	11.429	60.281	0.399	72.42	5.587	5.367	14.89 [13.49, 16.19]
6007	45.2	iota	3024	3056	5.484	8.342	0.325	14.76	9.480	5.790	4.66 [2.06, 10.60]
8009	46.9	full	8034	9242	17.741	224.328	0.358	241.48	18.915	10.964	22.33 [8.44, 38.29]
8009	46.9	iota	4018	4049	8.348	14.502	0.263	23.58	15.197	9.780	2.37 [1.96, 4.17]

Table 2: Medians of 3 seeds per row (`results/summary.csv`, from `results/attack.jsonl`). All 90 runs solved; rho solved all 90.

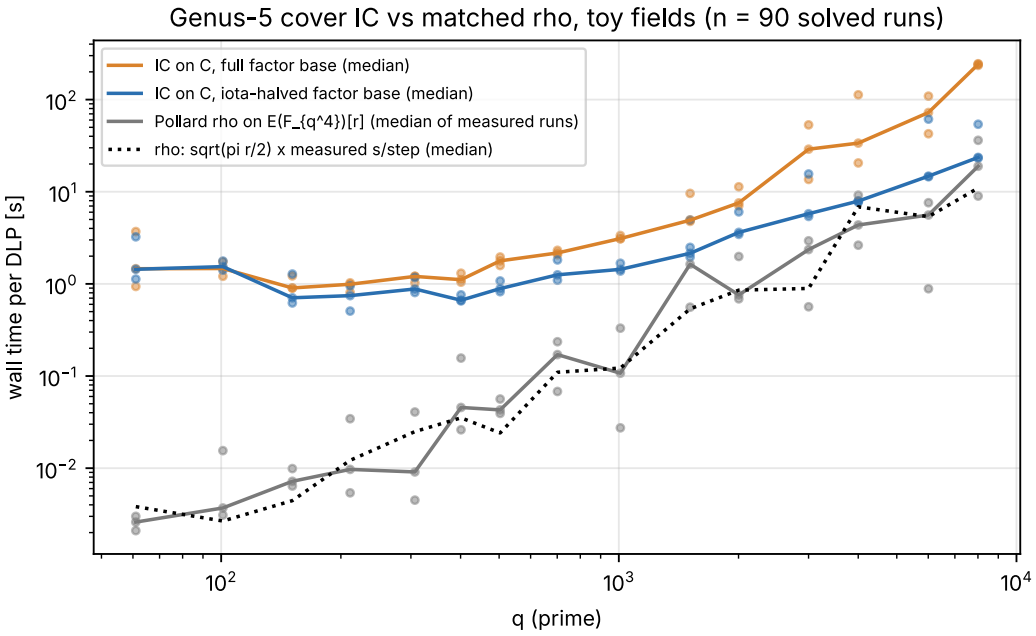


Figure 3: Wall time per DLP against q (log-log); points are single runs, lines medians.

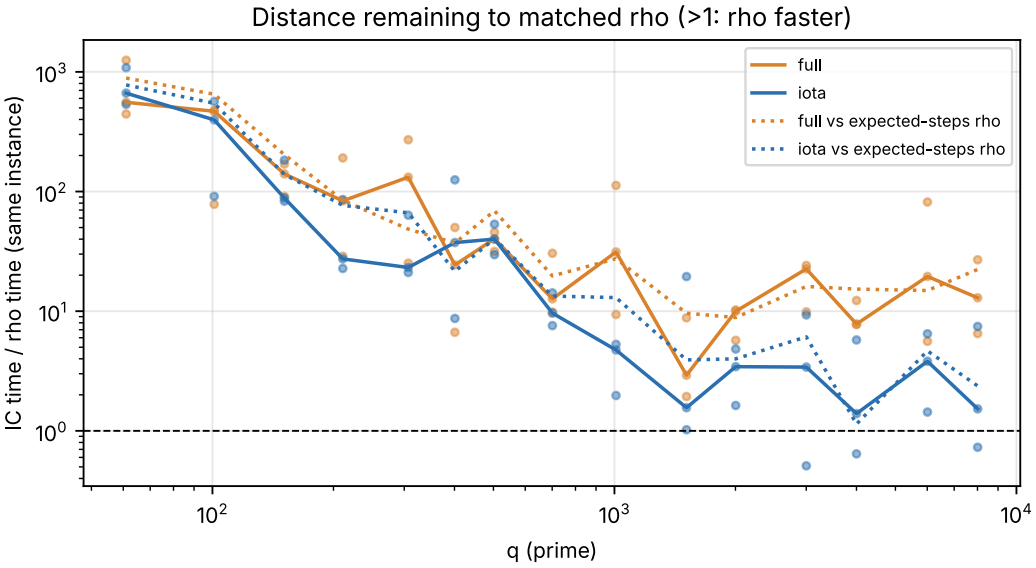


Figure 4: IC time over rho time on the same instance; above 1, rho is faster.

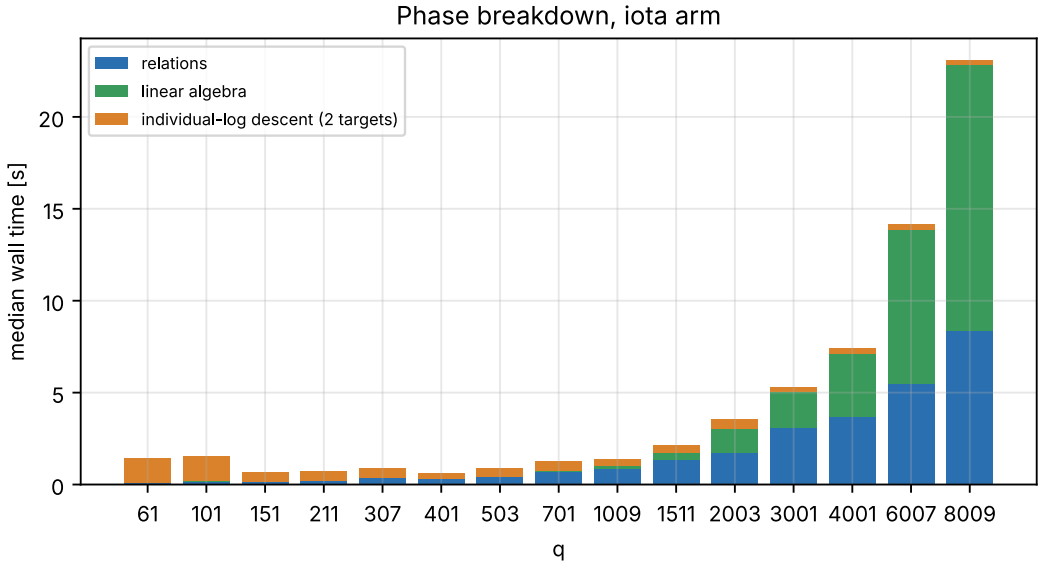


Figure 5: Median phase times, iota arm.

Observations (measured). Rho is faster at every q tested, by the median of each row and against the expected-steps reference in every single run. Three single runs fell below 1 against the **measured** rho run (iota arm: $q = 3001$ seed 3, 0.51×; $q = 4001$ seed 1, 0.64×; $q = 8009$ seed 1, 0.73×); in each, that rho walk took 1.8–2.7× its expected steps, so these are rho variance, not an IC advantage. Against the expected-steps rho, the iota arm is about 780× slower at $q = 61$ and between 1.14× ($q = 4001$) and 6.1× ($q = 3001$) slower for $q \geq 1511$, with no monotone trend in that range. The descent phase is roughly constant in q; relation search grows about linearly; sparse linear algebra grows faster and dominates the full arm at $q \geq 3001$. Peak RSS stayed below 14 MB.

Reading (not a claim beyond toy scale). Rho on $E(\mathbb{F}_{q^4})[r]$ costs about q^2 steps; this index calculus uses relations of weight 8 or 16 over a factor base of about q points, so its linear algebra is also about q^2. The remaining distance is therefore a constant factor in this regime. Large-prime variants that shrink the factor base are the standard route to a smaller exponent and were not implemented.

Pricing note (AGENTS.md §5a). Index calculus here performs no group operations on E; all of its work (norm evaluations, root finding, linear algebra) is unpriced native work. Priced at 0 GAE per operation it would count as zero group operations, and at 0.1 or 1 GAE it would count far more than rho's steps, so the order flips across the §5a band and the two arms are **not ranked on operation counts**. The only comparison reported is paired wall time on the same host, where rho is faster at every q.

4. Lead 4: the extra factor and covering symmetries

Derived (code comments in `src/cover.rs`). The involution $\iota : w \to -w$ is defined over $\mathbb{F}_q$, acts as -1 on the Res E part and $+1$ on D. In the r-part this forces $\log[\iota P] = c - \log[P]$ and $\log[D_\infty] = 4c$.

Verified on every run: both invariants hold for every factor-base pair (90/90).

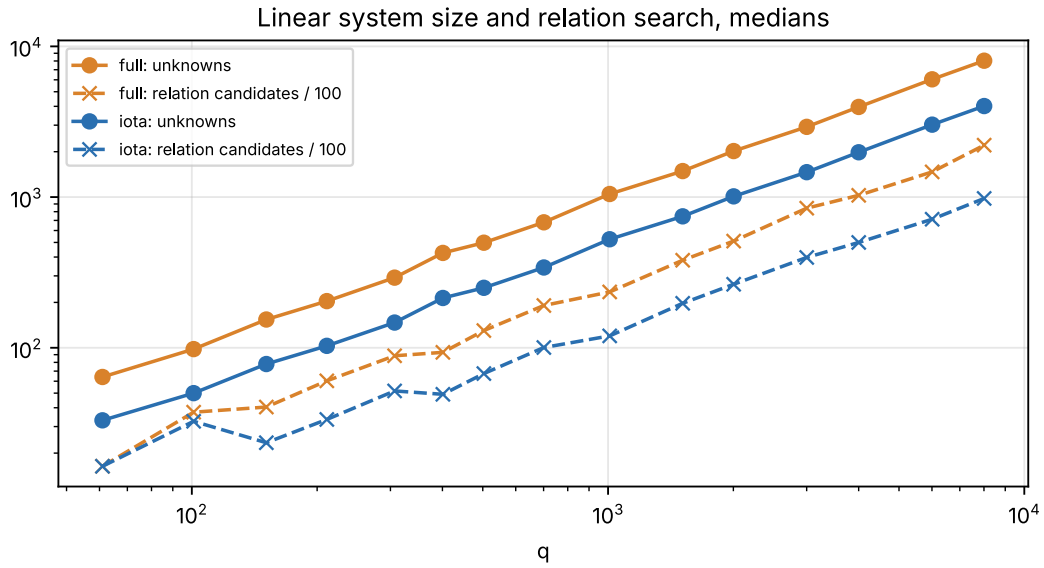


Figure 6: Unknowns and relation candidates, full vs iota arm (medians).

Measured, paired by instance (full / iota): unknowns 2.00× at every $q \geq 1009$; relation candidates about 2× (1.90–2.26× for $q \geq 401$); total IC time 1.95–10.2× for $q \geq 1009$ (10.2× at $q = 8009$, where linear algebra dominates). At small q the gain vanishes: at $q = 101$ the iota arm was slower (0.86×), because the descent phase, which iota does not shrink, dominates there (`results/summary.md`).

Other findings. Rational points of C lie over only about half of $x \in \mathbb{F}_q$, namely where $N(\eta)m_\alpha(x)$ is a square — the defining condition of D — so each residual root costs a density factor of about 1/2. No quotient of C over $\mathbb{F}_q$ other than the x-line has genus 0, so the cover yields no free relations. Hyperplane relations run out when $q^3 \lesssim 8!$; a quadric relation family is the fallback. Degree-8 places in the descent are avoided by descending $T(mP)$ for the smallest suitable m and dividing by m.

5. Failures and corrections

A first sweep (kept as `results/attack_v1_superseded.*`) returned one wrong logarithm ($q = 1511$, full arm): scalar Wiedemann gives a factor-base column that appears in no relation the value 0 under every column scaling, so an undetermined unknown went undetected. The fix adds an empty-column check and random column scaling to the corank test; the whole sweep was rerun with the fixed binary (90/90 correct). About 30 s of unit tests overlapped the rerun on the same host.

6. Limits

- Toy sizes only; no statement is made about cryptographic sizes.
- Wall time on a shared VM with no isolation; ratios are paired but not tier-qualified.
- Rho without the negation map; with it, rho would be about $\sqrt{2}$ faster.
- Appendix A cross-check limited to the quoted family.
- Conductor was not installed in this session.

7. Records

`results/verify.jsonl`, `results/classify.jsonl`, `results/attack.jsonl` (+ `.stderr`), `results/summary.csv`, `results/summary.md`, `results/host.txt`, `results/attack_v1_superseded.*`. Reproduce: `cargo test --release`; `./run_sweep.sh classify | verify | attack`; `python3 summarize.py`; `python3 plots.py`.